Testing Silicon Sampling in Public Opinion Research: Cross-National and Sociodemographic Patterns in LLM-Generated Survey Responses in the United States, Argentina, and Uruguay

2026-03-4

Abstract

1. Introduction

Large language models (LLMs), such as GPT-3 and GPT-4, have enabled new methods for generating human-like text. One recent approach, referred to as Silicon Sampling, involves the use of these models to simulate responses to survey questions. This method has been explored as a supplementary tool in social science research, particularly in settings where real data collection is costly, slow, or limited in scope. The growing interest in this technique reflects broader efforts in the field to explore computational methods that can support or augment traditional data collection processes. By producing text that mimics human responses, researchers have investigated whether LLMs can simulate representative views across demographic groups. This paper describes key applications of this technique, outlines methodological approaches, and highlights areas for further inquiry, particularly concerning validity, reliability, and the interpretive frameworks suitable for analyzing outputs generated by synthetic respondents.

2. Definition and Mechanisms of Silicon Sampling

Silicon Sampling refers to the practice of generating survey responses from LLMs by conditioning them on specified demographic or psychographic profiles. These profiles may include age, gender, ethnicity, geographic location, political affiliation, or other relevant attributes. The goal is to approximate how individuals with those traits might respond to survey items.

Sampling strategies vary in structure and intent. Random silicon sampling (RSS) involves drawing respondent profiles from real-world demographic distributions, such as national census data, and using those profiles to prompt the model. Stratified or quota-based approaches ensure that synthetic samples include fixed proportions of key subgroups, mirroring common survey design practices. These strategies aim to improve representativeness within synthetic samples, although they rely on assumptions about how prompts influence LLM outputs.

Other implementations adopt a purposive design, creating personas to examine specific subpopulations or scenarios rather than statistical representativeness. This approach is often used to explore how particular categories of individuals—such as undecided voters or patients with chronic conditions—might respond to different questions or messages. In some cases, researchers employ iterative sampling and filtering techniques, where multiple outputs are generated for a single profile, and only those meeting specific criteria are retained.

The process typically involves prompting the model to "act as" a respondent with given characteristics (e.g., "a 35-year-old Latino male living in Texas who voted for Biden"). These prompts are crafted to encourage consistent and contextually appropriate responses. Some studies use fixed persona libraries across survey items, while others dynamically adjust prompts or personas as the study progresses. Hybrid designs, which combine synthetic and human data, have also been introduced, allowing researchers to compare patterns across data types or use synthetic samples to pretest instruments.

3. Applications of Silicon Sampling

Silicon Sampling has been applied in several areas of social science research. In political science, studies have evaluated whether LLM-generated responses can approximate patterns found in traditional survey data. For instance, Argyle et al. (2022) used GPT-3 to generate responses to the American National Election Survey (ANES) and compared them with observed survey results. Their analysis focused on alignment at both aggregate and subgroup levels. Sun et al. (2024) used random silicon sampling to replicate distributions from the 2020 U.S. presidential election, reporting that model outputs were broadly similar to public opinion data along several demographic dimensions. Geng et al. (2024) explored the influence of prompt design in shaping synthetic responses, highlighting that outcomes varied significantly with seemingly minor changes in language or structure.

In policy research, synthetic samples have been used to simulate responses to policy proposals such as Universal Basic Income, climate regulation, and immigration reform. These studies typically use controlled prompts to test the effects of different framings or contextual cues on support or opposition. In some cases, researchers explore within-group variation by generating multiple responses from personas with similar demographic characteristics but slightly altered prompts.

In the domain of consumer behavior and marketing, synthetic personas have been used to explore product preferences, brand perception, and response to advertising content. Sarstedt et al. (2024) investigated the plausibility of LLM-generated consumer feedback, finding some correspondence with known patterns in human response data. These applications often serve as exploratory tools in early-stage product testing or messaging development.

In educational settings, synthetic respondents have been used to model hypothetical student feedback on instructional materials, simulate responses to curriculum changes, and examine engagement across different learner profiles. This approach is used primarily for exploratory purposes or to assess system responsiveness in automated tutoring environments.

Public health studies have applied silicon sampling to investigate responses to messaging around vaccination, mental health stigma, and health service delivery. For example, synthetic cohorts have been used to test reactions to vaccine information campaigns or alternative phrasing in mental health questionnaires. These studies often focus on content framing and comprehension.

Ethics and governance research has used synthetic personas in deliberative simulations and stakeholder exercises. These studies aim to understand how different kinds of individuals might reason through complex issues such as AI regulation or biomedical innovation. LLMs are prompted to simulate contributions in deliberative contexts, and the resulting outputs are analyzed thematically.

Qualitative research has employed LLM-generated text to simulate interview responses, stakeholder dialogues, and ethnographic vignettes. Amirova et al. (2024) compared LLM-generated interviews with transcripts from real interviews in UK health services. While some thematic alignment was noted, differences in tone, spontaneity, and the depth of narrative structure were also observed. These findings highlight both the potential utility and limitations of synthetic data in qualitative analysis.

Despite this growing literature, we still know less about whether LLM-generated survey responses can reproduce not only aggregate distributions, but also cross-national differences and internal sociodemographic structure. This paper addresses that gap by comparing synthetic and survey-based responses on political interest in the United States, Argentina, and Uruguay.

4. Data, Design, and Analytical Strategy

4.1 Data source and substantive focus

We evaluate silicon sampling using data from the World Values Survey (WVS), Wave 7, a large cross-national survey that collects comparable data on values, attitudes, and beliefs across a wide range of countries. The WVS is widely used in comparative research because of its standardized questionnaires, extensive

geographic coverage, and well-documented sampling procedures, which make it a valuable benchmark for assessing patterns of public opinion across different social and national contexts. In addition to providing the empirical reference for comparison, the survey also supplies the sociodemographic variables used to construct the synthetic respondent profiles.

Our analysis focuses on Q199, an item measuring respondents' interest in politics. The question asks respondents how interested they are in politics and offers a four-point response scale ranging from "Not at all interested" to "Very interested." Political interest is a central concept in the study of public opinion and political behavior, often interpreted as an indicator of political engagement and attentiveness to public affairs. It has been extensively studied in the comparative literature and is known to vary both across countries and across social groups within countries.

This variable is particularly suitable for evaluating silicon sampling for several reasons. First, it represents a well-established attitudinal dimension, making it possible to compare synthetic outputs with a large body of existing empirical research. Second, the distribution of political interest typically exhibits substantive variation across countries, reflecting differences in political institutions, media environments, and levels of civic engagement. Third, political interest is also strongly associated with sociodemographic characteristics such as education, age, occupation, and social class, allowing us to assess not only whether language models approximate overall response distributions, but also whether they reproduce the internal social structure of public opinion.

To examine these issues, we focus on three national contexts: the United States, Argentina, and Uruguay. These countries were selected in order to capture variation in political and institutional settings while maintaining comparability in survey design and question wording. The United States represents a highly studied case in survey research and provides a useful reference point for evaluating synthetic responses. Argentina and Uruguay, in turn, offer two Latin American contexts with distinct political trajectories and public opinion dynamics. By comparing these three cases, we assess whether LLM-generated responses reproduce both cross-national differences in political interest and within-country patterns of sociodemographic differentiation.

4.2 Synthetic respondent design

The unit of analysis in the synthetic data generation process is a synthetic respondent profile, defined by a combination of sociodemographic attributes embedded in the prompt provided to the language model. Each synthetic respondent is represented as a profile with specific demographic and social characteristics.

Synthetic profiles were constructed using a set of core attributes derived from the WVS: country, age, sex, education, occupation, and subjective social class. Each profile was assigned to one of the three national contexts examined in this study—the United States, Argentina, or Uruguay—and combined with values for the remaining sociodemographic variables. These attributes were chosen because they represent commonly used dimensions of social differentiation in survey research and are frequently associated with variation in political attitudes and engagement.

To guide the selection of attributes included in the synthetic respondent profiles, we conducted an exploratory analysis using the WVS dataset. Specifically, we estimated a Random Forest model to examine the relative importance of several sociodemographic variables in explaining variation in political attitudes, including interest in politics (Q199), evaluations of different political systems (Q235–Q239), and left–right ideological self-placement (Q240). Variable importance measures were used to identify which attributes contributed most strongly to explaining variation in these outcomes. The resulting importance plots (see Appendix Figure XX) indicate that country, age, education, occupation, and subjective class consistently appear among the most influential predictors, while other variables display weaker or less consistent contributions across outcomes.

Based on these exploratory results, we selected a parsimonious set of sociodemographic attributes—country, age, sex, education, occupation, and subjective class—to define the synthetic respondent profiles used in the prompting procedure. This choice allowed us to retain variables that are empirically associated with political attitudes while keeping prompts sufficiently concise to avoid excessive contextual complexity.

Each synthetic respondent profile was embedded into a structured prompt describing the individual's demographic and social characteristics. The prompt instructed the language model to answer the survey question as if it were a person with the specified attributes and answer the survey question from the perspective of a person with those attributes. By systematically varying these attributes across profiles, the design enables us to evaluate whether LLM-generated responses reproduce not only aggregate distributions but also structured variation across sociodemographic groups.

4.3 Models and prompting procedure

Synthetic responses were generated using several large language models (LLMs) that differ in architecture, training data, and parameter size. Including models with different characteristics allows us to evaluate whether patterns observed in silicon sampling are consistent across model families or whether they vary depending on model scale and design.

Table XX summarizes the models used in the analysis together with their approximate parameter size and the size categories employed.

Table 1: Large language models used in the analysis, model type, approximate parameter size, and size category.

Model	Type	Model size	Category
DeepSeek-R1	Open-weight	1.5B	Small
Qwen 3	Open-weight	1.7B	Small
llama3.2	Open-weight	3B	Small
gemma3	Open-weight	4B	Small
Llama 3.1	Open-weight	8B	Small
GPT-4.1 Mini	Proprietary	~15B	Medium
Gemini 2.5 Flash	Proprietary	~25B	Medium
gemma3	Open-weight	27B	Medium
Qwen 3	Open-weight	30B	Medium
GPT-4.1	Proprietary	~175B	Large

The models included represent both open-weight models and proprietary systems, covering a wide range of parameter sizes. To facilitate comparison, models were grouped into three broad categories according to their scale: Small (up to 10B parameters), Medium (10 to 30B parameters), and Large (more than 30B parameters). This categorization allows us to explore whether the ability of LLMs to reproduce survey response patterns varies with model capacity.

In general terms, larger models are expected to capture more complex statistical regularities from their training data, which could potentially improve their ability to approximate patterns observed in real survey responses. However, the relationship between model size and the sociological plausibility of generated responses remains an open empirical question. By comparing models across these size categories, the analysis allows us to examine whether increases in model scale are associated with improvements in the reproduction of aggregate distributions and sociodemographic patterns.

For each synthetic respondent profile, the model was prompted to answer the survey question while adopting the perspective of the specified individual. The prompt included a short description of the respondent's demographic and social characteristics (country, age, sex, education, occupation, and subjective social class) and instructed the model to answer the survey question as if it were a person with those attributes.

The prompt structure consisted of three elements. First, a brief description of the synthetic respondent profile was provided. Second, the survey question was presented using wording consistent with the original WVS item. Third, the set of valid response options was explicitly included in the prompt to constrain the model's output to the same response categories used in the survey.

Responses were generated independently for each model and respondent profile. By keeping the prompt structure constant across models, the design allows us to compare the resulting outputs while minimizing variation introduced by prompt design. The resulting dataset therefore consists of LLM-generated survey responses associated with synthetic respondent profiles, which can be analyzed using the same procedures commonly applied to survey data.

This procedure produces a dataset of synthetic responses associated with defined respondent profiles. Because the structure of the prompts and response options mirrors the original survey item, the resulting outputs can be analyzed using the same descriptive and statistical techniques commonly applied to survey data. This enables a direct comparison between synthetic and observed responses at both the aggregate level and across sociodemographic subgroups.

4.4 Analytical strategy

To evaluate the performance of silicon sampling, we compare synthetic and observed survey responses along three complementary analytical dimensions: aggregate correspondence, cross-national ordering, and sociodemographic structure.

First, we examine aggregate correspondence by comparing the overall distribution of synthetic responses with the empirical distribution observed in the WVS. This comparison allows us to assess whether language models reproduce the general shape of the response distribution or instead concentrate responses in particular categories.

Second, we analyze cross-national differences by examining whether models reproduce the relative ordering of countries in terms of political interest. This comparison evaluates whether synthetic responses capture meaningful variation across national contexts rather than simply approximating a global distribution.

Third, we examine sociodemographic structure by analyzing whether synthetic responses vary systematically across key social attributes such as age, education, occupation, and subjective class. This constitutes a more demanding test of silicon sampling, since it evaluates whether models reproduce the patterned internal differentiation commonly observed in survey data.

Together, these analytical dimensions allow us to distinguish between models that approximate aggregate response frequencies and those that also reproduce the relational and structural properties of observed public opinion data.

5. Results

5.1 Aggregate response distributions and variability

We begin by examining the overall distribution of responses to the political interest question in the WVS and comparing these distributions with those generated by the language models.

A primary characteristic of the simulations performed is the systematic under representation of the extremes of the scales. This is particularly evident in the case of the category "Very interested" which appears as a meaningful proportion (>10%) in only one model (gemini-2.5-flash-Medium), with most models showing either complete absence or minimal presence (<5%) of this response option across all geographic contexts.

The negative extreme of this scale -"Not at all interested"- appears as the dominant response only in qwen3-Medium (where it indicates severe mode collapse), while "Not very interested" and "Somewhat interested" serve as the preferred modal categories for most models.

Three groups of models appear to be observable: models with balanced multi-category distributions (e.g., deepseek-r1-Small showing approximately 50-60% in the modal category across all countries), models with extreme concentration (e.g., qwen3-Medium displaying 95% "Not at all interested" responses in Argentina versus a more balanced 44% in the USA), and models with moderate concentration that remains stable

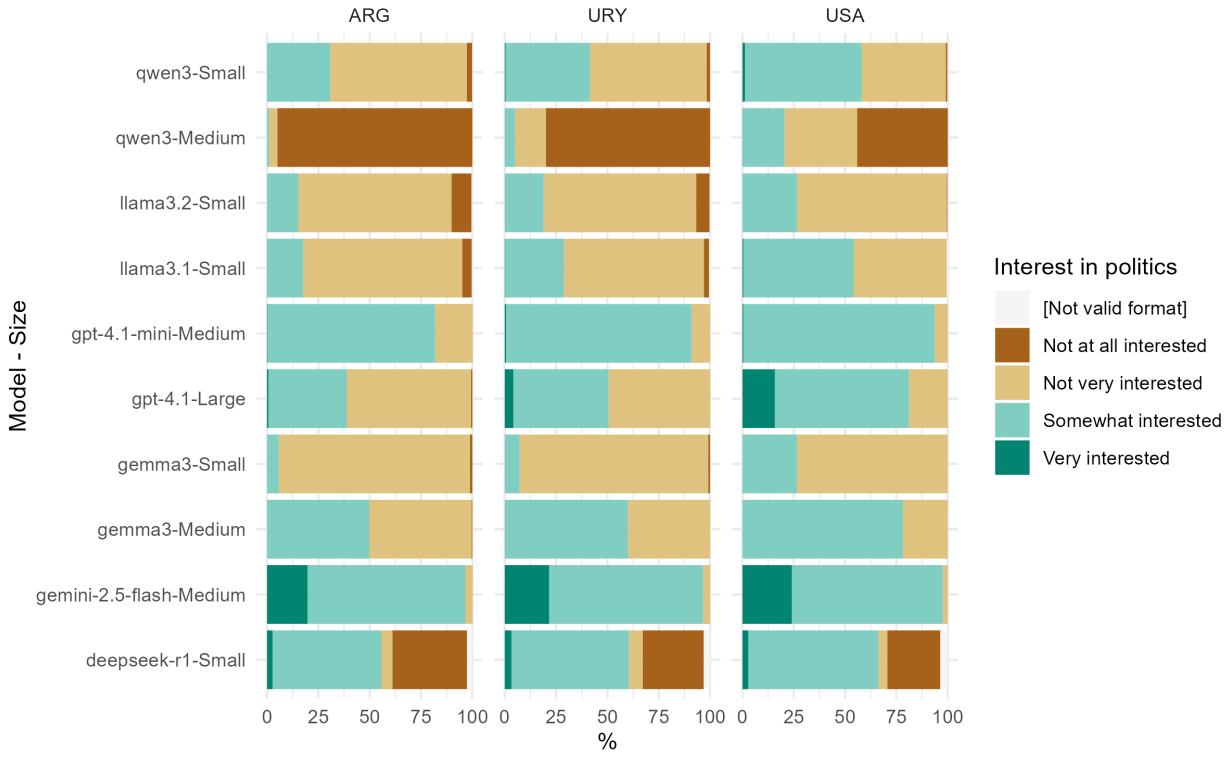

Figure 1: "Response distribution by model and country. Each bar shows the proportion of responses in five categories: 'Very interested' (dark teal), 'Somewhat interested' (light teal), 'Not very interested' (tan), 'Not at all interested' (dark brown), and invalid formats. Models are ordered by family and size. Note the extreme concentration in qwen3-Medium (ARG: 95% dark brown) versus balanced distribution in deepseek-r1-Small. The Uruguay advantage is evident in the higher proportion of light teal ('Somewhat interested') in the middle column compared to the left column for most models."

across contexts (e.g., llama3.2-Small maintaining approximately 73-75% in "Not very interested" across all countries).

Uruguay, seems to have somekind of advantage: it shows consistently higher proportions of "Somewhat interested" responses in the middle column compared to Argentina, with seven of nine models showing increased color diversity in Uruguay.

Different variability metrics confirm the previous analysis, since it reveals heterogeneity in response variability across language models and countries. Model-specific behavioral patterns emerge clearly: gpt-4.1-mini-Medium exhibits extreme response concentration across all countries (normalized entropy 0.16-0.31, modal percentages 81-93%), while deepseek-r1-Small maintains consistently high variability (entropy 0.62-0.67, effective categories 2.12-2.41).

However, significant geographic interactions complicate this picture. qwen3-Medium demonstrates dramatic cross-country behavioral shifts, operating near-deterministically in Argentina (entropy 0.14) while exhibiting the highest U.S. variability (entropy 0.66, effective categories 2.76). The Latin American samples display more pronounced bimodal distributions with clusters at behavioral extremes, whereas the U.S. sample shows smoother gradation across the variability spectrum.

These findings suggest that model response distributions reflect not merely architectural properties but complex interactions between training data composition, prompt interpretation mechanisms, and culturally-embedded survey content, raising fundamental questions about the portability of synthetic survey methodologies across demographic and linguistic boundaries.

Figure XX reports the Mean Absolute Error (MAE) between the distribution of synthetic responses generated by each model and the empirical distribution observed in the WVS for each country. Lower values indicate closer aggregate correspondence with the data.

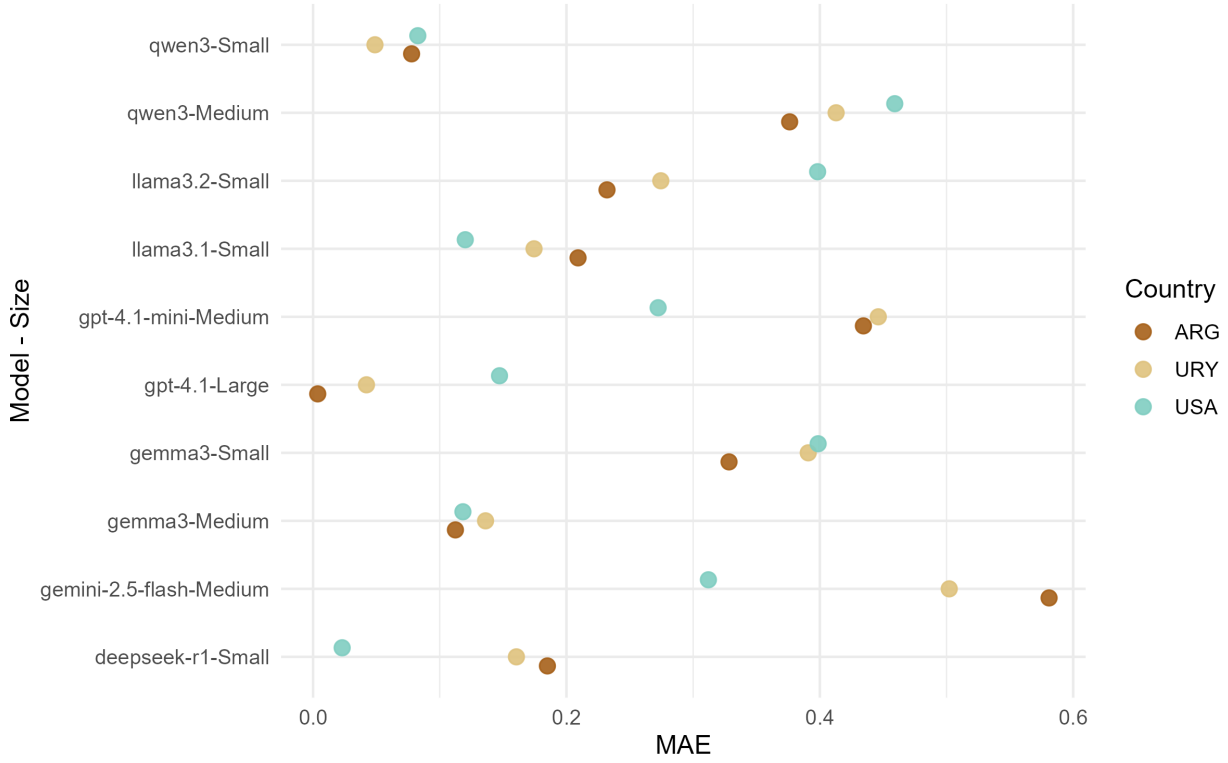

Figure 2: "Mean Absolute Error between the distribution of synthetic responses generated by each model and the empirical distribution observed in the WVS, by country. Each point represents the average absolute difference in response proportions across the survey categories. Lower values indicate closer correspondence between synthetic and observed distributions. Models are ordered by family and size, note the comparatively low error values for gpt-4.1-large and qwen3-small, and the substantially higher error levels for qwen3-Medium and gemma3-small across countries."

Among the best performing cases are qwen3-small in the United States, gpt-4.1-large in Argentina and Uruguay, and qwen3-small in Uruguay, all of which exhibit relatively low MAE values, indicating a close approximation to the empirical response distributions. At the opposite extreme, qwen3-Medium shows consistently high error levels across all three countries, reflecting strong concentration of responses in a single category. Similarly, gemma3-Small displays comparatively large deviations from the empirical distribution in each national context. These patterns are consistent with the response concentration and modal collapse dynamics observed in the previous analysis.

Systematic differences across countries within the same model are also evident. In several cases, the same model produces noticeably different error levels across Argentina, Uruguay, and the United States, suggesting that aggregate fit depends not only on model architecture or parameter size but also on the interaction between the model and the national response distribution represented in the prompt.

5.3 Subgroup structure

Con el objetivo de evaluar si los modelos de lenguaje reproducen no solo las distribuciones agregadas de respuesta, como veíamos en el MAE, sino también la estructura social de la opinión, estimamos modelos de regresión logística separados para cada uno, es decir WVS y cada uno de los LLMs utilizados, usando las

mismas variables sociodemográficas como predictores del interés político considerado de forma dicotomica. Lo que permite la regresión es visualizar no solo las proporciones similares en términos de promedio, sino conocer cuánto cambia la probabilidad de la variable dependiente, interés político, cuando cambia cada atributo, controlando por los demás,

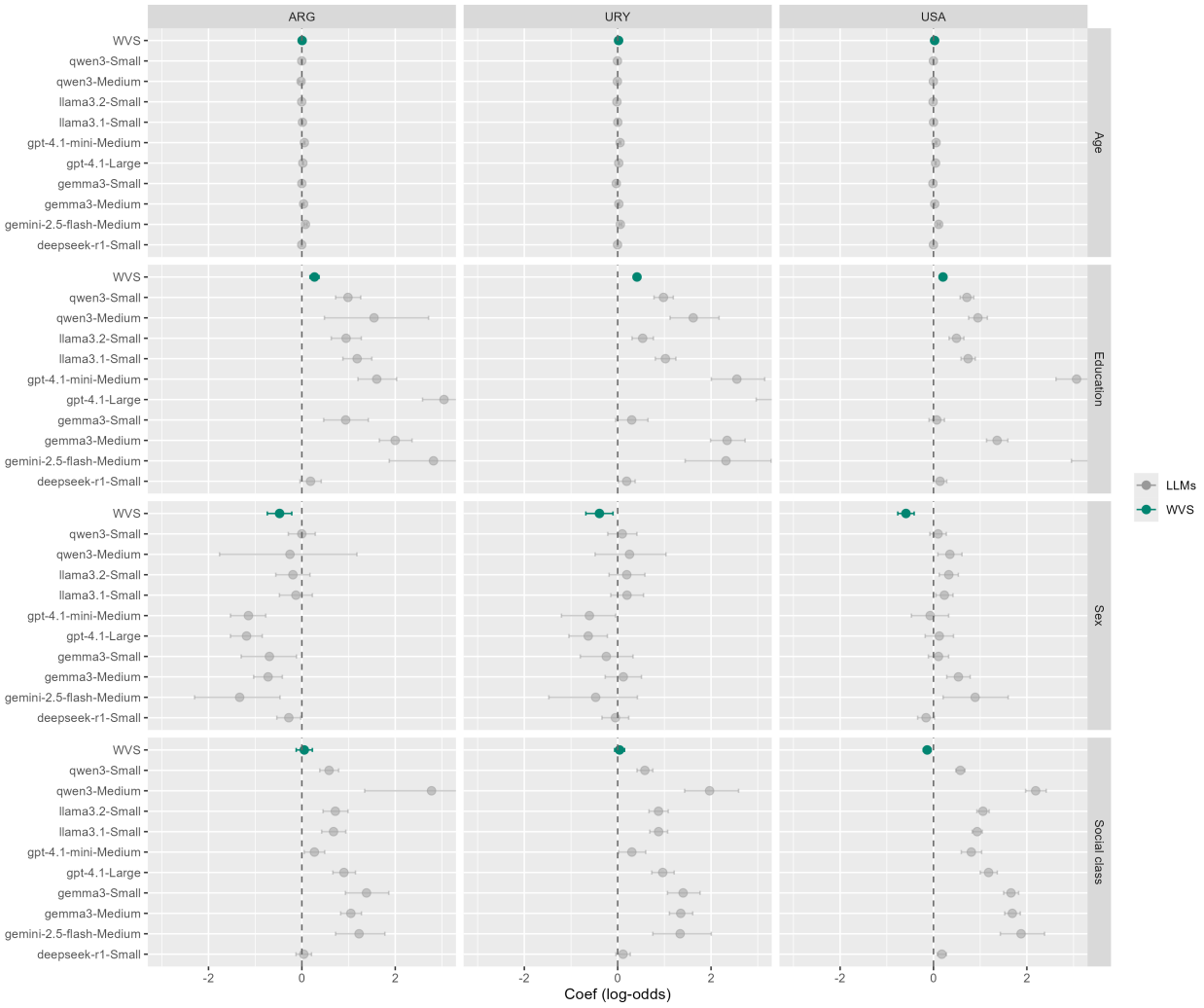

Figure 3: "Los coeficientes de regresión predicen el interés político (dicotómico) por modelo y país. Cada punto representa el coeficiente estimado para un predictor sociodemográfico dado, con líneas horizontales que indican intervalos de confianza al 95%. El punto verde corresponde a la línea de base empírica del WVS y los puntos grises corresponden a cada LLMs. Los modelos están ordenados por familia y tamaño. Se destaca la sobreestimación sistemática de los efectos de educación y clase social en todos los modelos, los coeficientes de edad cercanos a cero en relación con la línea de base del WVS, y el signo inconsistente del coeficiente de sexo entre todos los modelos considerados. La mayor dispersión de los coeficientes de LLMs en Argentina y Uruguay, en comparación con Estados Unidos, es particularmente evidente en las variables sexo y clase social."

Se observa que el efecto de la edad en los datos reales de la WVS es minimamente positivo en los tres países, ya que a mayor edad existe algo más de interés político. Los LLMs en general aciertan en la dirección pero los coeficiente se acercan casi a cero, es decir, que no lo distorsionan en gran medida.

En el caso de la educación, los datos de la WVS muestra un efecto positivo en los tres países, por lo que a más educación, más interés en la política, lo cual es consistente con la literatura sobre participación política.

Sin embargo, se visualiza que todos los LLMs estudiados sobreestiman ese efecto de manera sistemática, en algunos casos muy por encima del WVS. La dirección es es correcta pero la magnitud está sobreestimada. También existe una dispersión considerable entre modelos, especialmente en Argentina

Con respecto a la variable sexo, la WVS muestra un efecto negativo consistente en los tres países, es decir, las mujeres tienen menos interés político que los hombres. Los LLMs analizados fallan considerablemente en este aspecto ya que hay modelos que invierten el signo, otros que lo sobrestiman, y los intervalos de confianza son amplios en general. Hay además una diferencia geográfica relevante, la dispersión entre modelos es mayor en Argentina y Uruguay que en Estados Unidos, lo que sugiere que los modelos tienen menos "certeza" sobre cómo se expresa las diferencias por sexo en contextos latinoamericanos.

Sobre la clase social, en la WVS esta variabletiene un efecto positivo pequeño, la clase más alta presenta algo más de interés político, pero en poca magnitud. En estos casos, los LLMs sobreestiman este atributo consistentemente, y en Estados Unidos de forma particularmente acentuada.Esto implica un representación subyacente de la clase social como predictor de interés político, mucho más pronunciada que lo que muestran los datos reales.

En suma, el patrón general es que los LLMs capturan razonablemente bien la dirección de los gradientes sociodemográficos considerados pero sistemáticamente exageran su magnitud, especialmente para educación y clase social. Esto refuerza la idea de existencia de determinados estereotipos sociales sobre quién está interesado en política y cuan relevante son sus características. El sexo es el más problemático porque ni siquiera la dirección es consistente entre modelos, y se observa que las distorsiones son más acentuadas en los contextos latinoamericanos que en Estados Unidos.

Conclusions

Appendix

Table A1: Modal Response Concentration (%) by Model and Country

Model and size	Modal cat. percent._ARG	Modal cat. percent._URY	Modal cat. percent._USA	Modal response_ARG	Modal response_URY	Modal response_USA
deepseek-r1-Small	52.84	57.2	63.37	Somewhat interested	Somewhat interested	Somewhat interested
gemini-2.5-flash-Medium	76.87	75.0	73.54	Somewhat interested	Somewhat interested	Somewhat interested
gemma3-Medium	49.85	60.0	78.24	Not very interested	Somewhat interested	Somewhat interested
gemma3-Small	93.12	91.8	73.46	Not very interested	Not very interested	Not very interested
gpt-4.1-Large	60.42	49.3	65.33	Not very interested	Not very interested	Somewhat interested
gpt-4.1-mini-Medium	81.75	90.4	93.22	Somewhat interested	Somewhat interested	Somewhat interested
llama3.1-Small	77.57	68.2	53.70	Not very interested	Not very interested	Somewhat interested
llama3.2-Small	74.58	74.3	72.96	Not very interested	Not very interested	Not very interested
qwen3-Medium	94.92	80.0	44.18	Not at all interested	Not at all interested	Not at all interested

Model and size	Modal cat. percent._ARG	Modal cat. percent._URY	Modal cat. percent._USA	Modal response_ARG	Modal response_URY	Modal response_USA
qwen3-Small	66.60	56.8	56.82	Not very interested	Not very interested	Somewhat interested